

Effects of Al addition on structural evolution and tensile properties of the FeCoNiCrMn high-entropy alloy system

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Received 18 July 2013; received in revised form 19 September 2013; accepted 22 September 2013

Available online 16 October 2013

Abstract

A series of six-component (FeCoNiCrMn)_{100-x}Al_x ($x = 0-20$ at.%) high-entropy alloys (HEAs) was synthesized to investigate the alloying effect of Al on the structure and tensile properties. The microstructures of these alloys were examined using transmission electron microscopy, and crystalline phase evolution was characterized and compared with existing models. It was found that the crystalline structure changed from the initial single face-centered cubic (fcc) structure to a duplex fcc plus body-centered cubic (bcc) structure and then a single bcc structure as the Al concentration was increased. Resulting from the structural changes there were also corresponding variations in tensile properties. In the single fcc region, alloys behaved like a solid solution with relatively low strength but extended ductility. In the mixed structure region, alloys behaved like a composite with a sharp increase in strength but reduced ductility. In the single bcc region, alloys became extremely brittle. In this study, close correlation between the microstructure and mechanical properties was also discussed and presented.

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Keywords: High-entropy alloys; Microstructures; Mechanical properties; Phase transition; Solid solution

1. Introduction

In the past decade, high-entropy alloys (HEAs) have attracted extensive research attention because of their unusual structural properties [1–5]. The traditional design of alloys involves selecting the major component based on a specific property requirement, and further using alloying additions to confer secondary properties without sacrificing the primary property. By contrast, HEAs are multicomponent alloys containing several components (usually ≥ 5) in approximately equiatomic proportions. However, it is of particular interest to note that, despite containing a large number of components, HEAs actually exhibit a significant degree of mutual solubility and tend to form simple face-centered cubic (fcc) and/or body-centered cubic (bcc) solid

solutions, instead of complex phases and intermetallic compounds [6], which is often attributed to a high mixing (or configuration) entropy.

Several studies showed that additions of certain elements into HEAs could have strong effects on the crystalline structure, microstructural morphology and the subsequent mechanical properties [7–15]. In the Al_xCoCrFeNiTi system [7], for example, the as-cast structure tends to evolve from a mixture of fcc + bcc duplex phases to a single bcc phase with the increase of Al concentration. In another AlCoCrFeNb_xNi system [8], it was found that increasing the addition of Nb promoted the transition from hypoeutectic to hypereutectic and such transition had a pronounced effect on the compressive yielding strength and Vickers hardness. However, quantitative and systematic studies on the alloying effects on the microstructural evolution, particularly the transition from one solid solution phase to another, are still rare. In addition, as a result of low ductility and brittleness in many

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HEAs, most of the reported mechanical properties were either obtained from compression tests or derived from hardness measurements due to a lack of direct information on plastic instability. Comprehensive understanding of the alloying effects on the microstructural and phase evolution, and their impacts on the subsequent tensile properties and strengthening mechanism, are necessary for further development of this new class of materials for structural applications.

In this study, we selected a FeCoNiCrMn alloy (all elements in equal atomic proportions) as our baseline alloy system. This alloy has a single fcc phase structure [16] and large compressive plasticity. A systematic study of alloying effects of Al on crystal structure, phase evolution, hardness and tensile properties of the (FeCoNiCrMn)_{100-x}Al_x ($x = 0–20$ at.%) alloys were conducted. The phase formation mechanism and the correlation between property and microstructure were presented.

2. Experimental

Alloy ingots with a nominal composition of (FeCoNiCrMn)_{100-x}Al_x ($x = 0, 2, 3, 4, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16$ and 20 at.%) were synthesized by arc-melting a mixture of pure metals (purity >99 wt.%) in a Ti-gettered high-purity argon atmosphere. These ingots were remelted at least four times to ensure chemical homogeneity. Melted alloys were eventually drop-cast into a mold $10\text{ mm} \times 10\text{ mm} \times 60\text{ mm}$.

Phase identification of all the specimens was conducted by X-ray diffraction (XRD) using Cu K_α radiation (MXP21VAHF). The morphology and microstructure of the as-cast specimens were characterized by scanning electron microscopy (SEM) using a Zeiss Supra55 operated at 15 kV with an energy dispersive spectrometer (EDS), and by high-resolution transmission electron microscopy (TEM) with a JEM-2010. For microstructural observation, samples were sequentially ground and polished down to $0.1\text{ }\mu\text{m}$ diamond polishing compound, and etched with a solution composed of hydrochloric acid, sulfuric acid and nitric acid (in a volume ratio of 75:10:5) as well as supersaturated copper sulfate. TEM samples were first mechanically ground to $\sim 40\text{ }\mu\text{m}$ thick and then twin-jet electropolished using a mixed solution of $\text{HNO}_3:\text{CH}_3\text{OH} = 1:4$. Volume fractions of different phases in the alloys were estimated from ten individual areas using the grid method. Tensile samples with a gauge length of 15 mm, a width of 5 mm and an approximate thickness of 1.3 mm were prepared via electric discharging machining. Tensile tests were carried out on a CMT4105 universal electronic tensile testing machine with a strain rate of $1 \times 10^{-3}\text{ s}^{-1}$ at room temperature. Hardness measurements were conducted using a Vickers hardness tester (430SVD) with a load of 5 N for 15 s, and for each specimen, at least 15 indents were measured to obtain an average value.

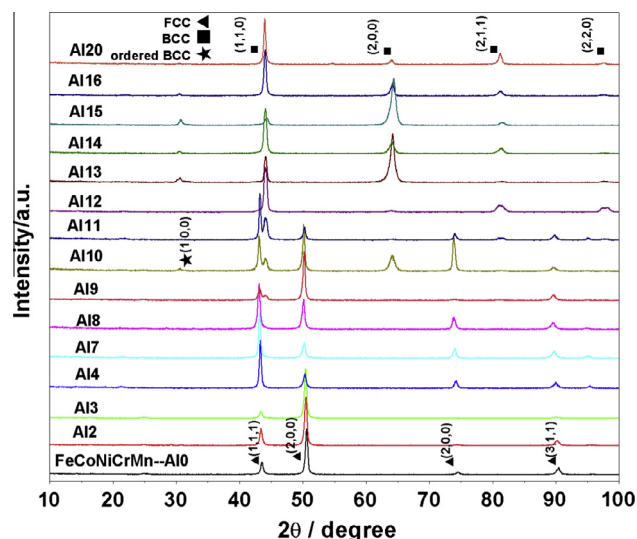


Fig. 1. XRD patterns from the as-cast (FeCoNiCrMn)_{100-x}Al_x (Al_x) samples indicate a phase transition from fcc to bcc starting at $\sim \text{Al}_8$.

3. Results

3.1. Structural evolution as a function of the Al concentration

XRD patterns from the as-cast (FeCoNiCrMn)_{100-x}Al_x alloys (denoted as Al_x hereafter) are shown in Fig. 1. Apparently, there is a crystal structure transition from fcc to bcc as the Al content is increased. When the Al concentration is less than 8% (i.e. $x < 8$), only a single fcc crystal structure is observed. However, the $(110)_{\text{bcc}}$ peak near the $(111)_{\text{fcc}}$ peak begins to appear in the alloy with 9% Al, along with a minor peak from the ordered bcc near $2\theta \approx 30^\circ$ identified as the (100) plane. In alloys containing a high Al content (i.e. $x \geq 16$), only bcc diffraction peaks are identified, suggesting that either the fcc phase no longer exists or its amount is below the detectable limit of XRD.

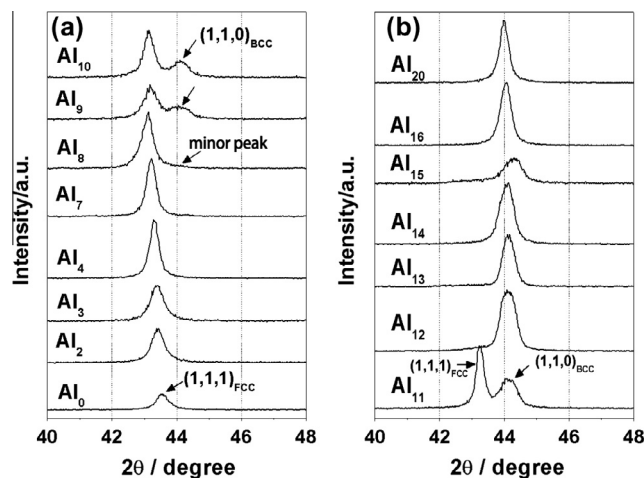


Fig. 2. Details of $(111)_{\text{fcc}}$ and $(110)_{\text{bcc}}$ peak shifting in as-cast Al_x alloys: (a) Al₀–Al₁₀, (b) Al₁₁–Al₂₀. A discernible minor peak arises in Al₈ at $2\theta \approx 40^\circ$, indicating the presence of a small volume of bcc phase.

measurements. Detailed evolution of the two characteristic crystalline peaks, i.e. $(111)_{\text{fcc}}$ and $(110)_{\text{bcc}}$, are demonstrated in Fig. 2a and b. As indicated, the $(111)_{\text{fcc}}$ peak initially shifts towards the low 2θ angle, but switches to the high 2θ angle in the alloys containing more than 8% Al. The $(110)_{\text{bcc}}$ peak shifting is more complicated in alloys having an Al content from 8% to 16%, as a result of the formation of duplex solid solutions. It is worth noting that, in alloy Al₈, a weak peak already commences at almost the same 2θ degree as the $(110)_{\text{bcc}}$ peak of alloy Al₉, suggesting the formation of a low volume fraction of the bcc phase.

To further investigate phase formation caused by the Al addition, microstructures of the as-cast alloys with different Al contents are presented in Fig. 3. Al₀ and Al₄ alloys are selected to represent alloys with a single fcc phase (see Fig. 3a and b). Equiaxed fcc grains are observed in these alloys, consistent with the XRD patterns shown in Fig. 1. Fig. 3c–j exhibits microstructural morphologies of the Al₈, Al₉, Al₁₀, Al₁₁, Al₁₂, Al₁₃, Al₁₄ and Al₁₆ samples, respectively. In Fig. 3c (i.e. Al₈), few dark regions (identified as the bcc phase) are found to be located along grain boundaries; these dark regions correspond to the minor peak mentioned in the XRD analysis (Fig. 1). As the Al content is further increased, the dark region also expanded. In Al₁₀, equiaxed grains are gradually replaced by bright

dendrites, and the dark bcc phase is spread in the inter-dendrite region (see Fig. 3d and e). Chemical compositions of the dendritic and inter-dendritic regions were analyzed using EDS and the results are listed in Table 1. The dark bcc phase is noted to contain much more Al than the bright fcc dendrites.

With additional Al, the morphology of the fcc phase further evolves from dendrites to a lamella-like structure known as the Widmanstätten side-plates [17], whilst the bcc phase constitutes the inter-side-plate areas in the Al₁₁ and Al₁₂ samples (Fig. 3f and g). Long side-plates developed from boundaries and grew across grains in Al₁₁, but in Al₁₂ alloy these plates are much shorter and thinner. This bcc phase gradually becomes dominant in Al₁₃, Al₁₄ and Al₁₆ (Fig. 3h–j), and the minor fcc phase mainly lies in the bcc grain boundaries. A nano-scale, basket-weave-like structure is found in the bcc structure (e.g. Al₁₄ and Al₁₆ shown in Fig. 3i and j). Combining with the XRD results, it is speculated that all the bcc phases were initially formed via spinodal decomposition into modulated plate structures of the disordered bcc (A2) phase and the ordered bcc (B2) phase [18,19]. A similar observation has also been reported before in other Al-containing HEA systems, e.g. Al_xCoCrCuFeNi and Al_xCoCrFeNi [20,21], which also offers a reason for the appearance of two image contrasts

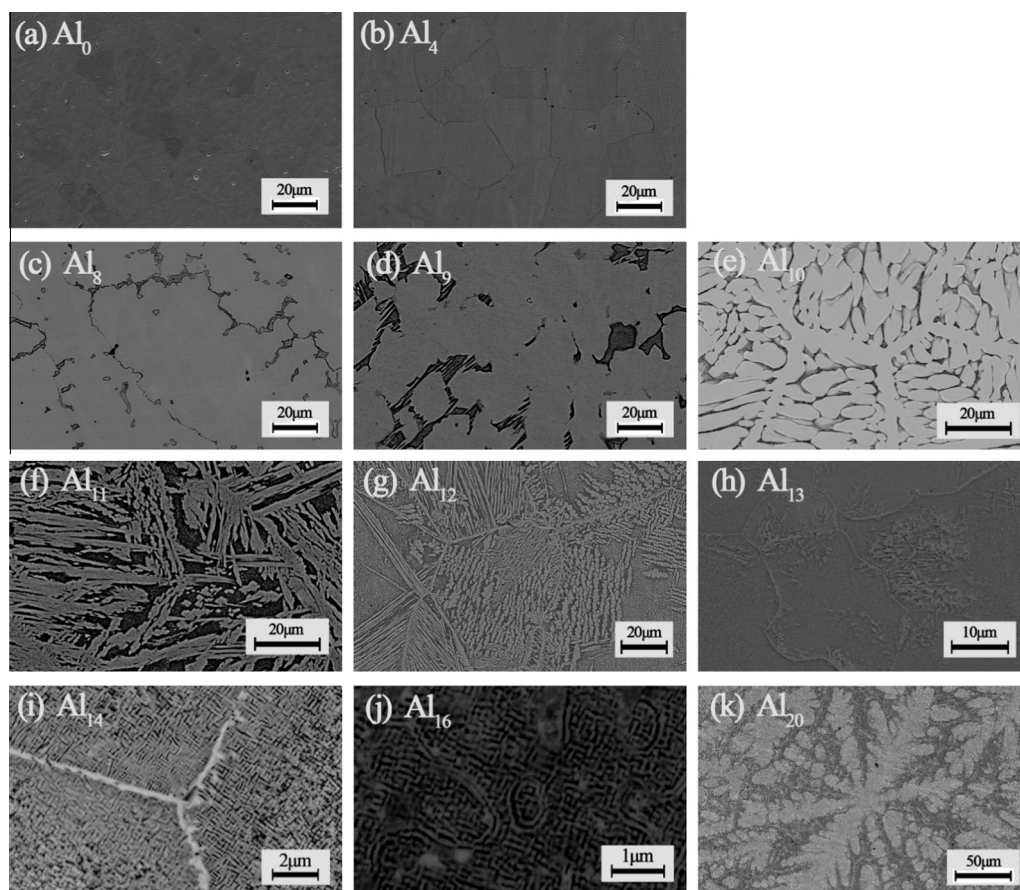


Fig. 3. SEM micrographs of etched as-cast Al_x alloys: (a) is a secondary-electron image of Al₀ and (b–k) are backscattered images for Al₄–Al₂₀, respectively.

Table 1
Chemical compositions of different regions in the (FeCoNiCrMn)_{100-x}Al_x high-entropy alloys by EDS (at.%).

Alloys	Regions ^a	Crystal structure	Chemical compositions/at. %					
			Al	Cr	Mn	Fe	Co	Ni
Al ₀	O	FCC	0.0	21.3 ± 0.4	20.7 ± 0.5	19.4 ± 0.5	19.3 ± 0.6	19.3 ± 0.7
Al ₂	O	FCC	2.4 ± 0.8	20.9 ± 0.4	19.7 ± 0.5	19.5 ± 0.5	19.3 ± 0.4	18.2 ± 0.6
Al ₄	O	FCC	4.6 ± 0.1	20.9 ± 0.3	18.1 ± 0.3	19.1 ± 0.4	19.3 ± 0.5	18.0 ± 0.5
Al ₈	O	–	7.6 ± 0.2	20.0 ± 0.3	18.2 ± 0.3	18.3 ± 0.4	18.2 ± 0.2	17.7 ± 0.5
	BP	FCC	7.0 ± 0.4	20.9 ± 0.2	15.4 ± 0.2	21.1 ± 0.3	19.4 ± 0.1	16.2 ± 0.4
	DP	BCC	11.2 ± 0.5	17.8 ± 0.3	22.7 ± 0.4	14.5 ± 0.2	15.2 ± 0.3	18.6 ± 0.2
Al ₉	O	–	9.3 ± 0.4	19 ± 0.5	18.4 ± 0.4	17.5 ± 1.1	18 ± 0.5	17.8 ± 0.8
	BP	FCC	8 ± 0.6	20.5 ± 0.4	16 ± 0.6	20 ± 0.4	19.1 ± 0.2	16.4 ± 0.3
	DP	BCC	14.1 ± 1.1	15.8 ± 1.5	21.1 ± 1.2	13.3 ± 0.6	15.2 ± 0.6	20.5 ± 0.4
Al ₁₀	O	–	10.4 ± 0.8	19 ± 0.4	18 ± 0.3	17.4 ± 0.5	17.6 ± 0.2	17.6 ± 0.2
	BP	FCC	9.1 ± 0.2	18.8 ± 0.5	17.9 ± 0.3	18.1 ± 0.4	18.2 ± 0.5	17.9 ± 0.4
	DP	BCC	14.7 ± 0.5	20 ± 1.0	17.1 ± 0.5	15.9 ± 0.5	16.2 ± 0.4	16.1 ± 0.4
Al ₁₁	O	–	11.9 ± 0.4	17.9 ± 0.8	17.9 ± 0.7	17.6 ± 0.7	17.1 ± 0.3	17.6 ± 0.5
	BP	FCC	10.9 ± 1.0	17.4 ± 0.6	18.3 ± 0.4	17.4 ± 0.1	17.6 ± 0.2	18.4 ± 0.6
	DP	BCC	13.3 ± 0.5	19.5 ± 0.8	17.1 ± 0.8	17.2 ± 0.3	16.9 ± 0.7	16 ± 1.2
Al ₁₂	O	–	13.1 ± 0.7	18.2 ± 0.4	17.3 ± 0.4	17.2 ± 0.2	17 ± 0.4	17.2 ± 0.2
Al ₁₃	O	–	14 ± 0.2	17.9 ± 0.7	17.2 ± 0.4	16.8 ± 0.3	16.9 ± 0.3	17.2 ± 0.4
Al ₁₄	O	–	14.4 ± 0.8	18.3 ± 0.5	16.9 ± 0.5	16.8 ± 0.2	16.9 ± 0.4	16.7 ± 0.2
Al ₁₆	O	–	17.5 ± 1.2	17.4 ± 0.4	16.4 ± 0.8	16.2 ± 0.4	16.1 ± 0.2	16.3 ± 0.3
Al ₂₀	O	BCC	21.3 ± 0.3	16.4 ± 0.4	15.9 ± 0.9	15.5 ± 0.2	15.4 ± 0.6	15.5 ± 0.5
	BP	BCC	20.5 ± 1.8	15.8 ± 0.2	13.9 ± 0.6	15.7 ± 0.4	16.9 ± 0.6	17.2 ± 0.3
	DP	BCC	24.9 ± 1.1	5.3 ± 0.4	17.3 ± 0.5	9.1 ± 0.5	20.1 ± 0.3	24.3 ± 0.3

^a O: overall, BP: bright phase, DP: dark phase.

in the bcc phase region. Finally, the morphology of the simple bcc alloy Al₂₀ is presented in Fig. 3k in which equiaxial dendrites are clearly revealed with distinctive chemical segregation, mainly Fe and Cr, as indicated in Table 1.

In summary, the current high-entropy (FeCoNiCrMn)_{100-x}Al_x alloys in the as-cast state can be divided into three categories based on their crystalline structures: single fcc alloys with less than 8% Al (i.e. $x < 8$), duplex fcc–bcc alloys with the Al content between 8% and 16% (i.e. $8\% < x < 16\%$) and simple bcc alloys with more than 16% Al (i.e. $x > 16\%$). For easy discussion, we herein denote these three categories as regions I, II and III, respectively.

3.2. Tensile properties and hardness

Engineering stress–strain curves for the current HEAs containing different amounts of Al are depicted in Fig. 4a. The fracture strength, yield strength and tensile ductility as a function of Al are displayed in Fig. 4b, whilst hardness vs. the Al content is also shown in Fig. 4c. It is noted that the hardness over tensile strength ratio is $\sim 3.13 \pm 0.39$ in the current alloys, similar to that in conventional steels [22], and obeys well the Tabor relationship [23]. In region I, both the fracture and yield strength, as well as hardness, slightly increase with increasing Al concentration. Specifically, the fracture and yield strength of Al₀ are 496 and 209 MPa, respectively, but increase slightly to 529 and 242 MPa, respectively, in Al₇; the corresponding hardness increases from 176 HV for Al₀ to 182 HV for Al₇.

Meanwhile, tensile elongation reduces from 61.7% in Al₀ to 47.2% in Al₇, but is still quite high.

The work hardening coefficient N can be calculated using the Holloman equation:

$$\sigma = K\varepsilon^N \quad (1)$$

where σ is the stress, ε is the plastic strain and K is a material constant [24]. The N values are calculated to be 0.56, 0.53, 0.50, 0.47, 0.4, 0.2 and 0.15 in Al₀, Al₄, Al₇, Al₈, Al₉, Al₁₀ and Al₁₁, respectively, which decreases with increasing Al concentration. The descending value with increasing Al content is noted to be consistent with the ductility trend exhibited in Fig. 4a, according to the classical Considère criterion for plastic instability, namely, $\varepsilon_{necking} \approx N$ [25]. In fact, plastic instability or sample necking was observed in the current alloys, especially when they are in a solid-solution state, i.e. Al_x ($x = 0-8$).

For the alloys in region II, in contrast to region I, both tensile and yield strength rise sharply with the Al content, and reach their maximum values of 1174 and 832 MPa, respectively, at 11% Al (or in the Al₁₁ alloy). However, tensile elongation also rapidly decreases with the increase of Al and reduces to 7.7% in Al₁₁. When the Al concentration exceeds 11%, the as-cast specimens become very brittle and cannot be stretched in tension. In parallel, their Vickers hardness value also sharply increases with the Al level and approaches a value of 538 HV in Al₁₄. In region III, alloys with more than 16% Al, the hardness value becomes saturated and remains essentially the same.

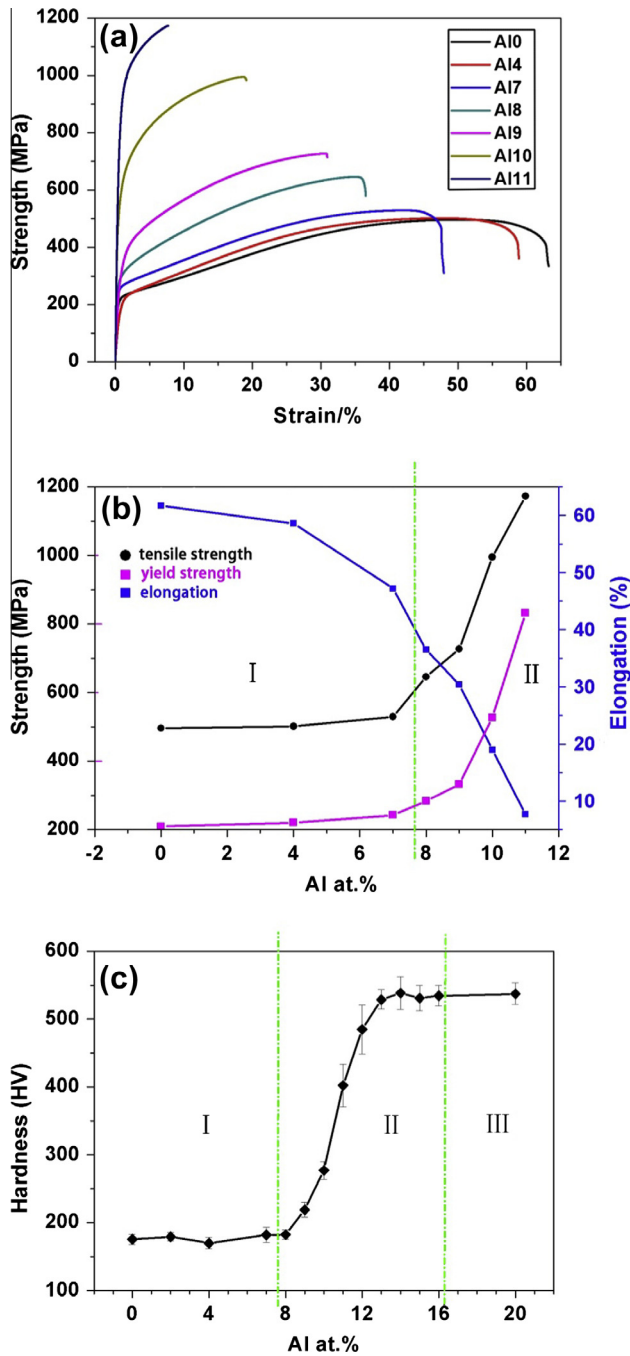


Fig. 4. Mechanical properties of the as-cast Al_x alloys at room temperature: (a) engineering stress–strain curves of Al_x , (b) tensile strength, yield strength and elongation as a function of Al and (c) Vickers hardness variation with the Al content.

4. Discussion

4.1. Phase formation and transition

In conventional alloys such as steels, the addition of aluminum usually promotes the formation of bcc α -Fe and destabilizes fcc γ -Fe [26]. In a similar manner, aluminum is also a strong bcc stabilizer and the phase transition from fcc to bcc induced by Al addition has also been observed in

many HEA systems [7,12,21,27]. However, systematic characterization of the microstructure–property relationship as a result of Al additions has not yet been performed.

Several physical parameters have been proposed to predict structural stability and phase formation in HEAs [28,30–32]. From a thermodynamic point of view, and in particular derived from Hume–Rothery rules, Zhang et al. suggested a criterion for achieving solid-solution phases in HEAs using two parameters, ΔH_{mix} (enthalpy of mixing) and δ (atomic radius difference) [28], as expressed below:

$$\delta = \sqrt{\sum_{i=1}^n c_i (1 - r_i/\bar{r})^2} \quad (2)$$

$$\Delta H_{\text{mix}} = \sum_{i=1, i \neq j}^n \Omega_{ij} c_i c_j = 4 \sum_{i=1, i \neq j}^n \Delta H_{ij}^{\text{mix}} c_i c_j \quad (3)$$

where r_i is the Goldschmidt atomic radius of the i th element, c_i is the molar ratio and $\Delta H_{ij}^{\text{mix}}$ is the enthalpy of mixing between the i th and j th elements. The parameter δ characterizes the atomic size mismatch, which produces the local elastic strain and determines the system topological instability [29], while the parameter ΔH_{mix} reflects the tendency of forming stable intermetallic compounds. Using available data from various HEAs systems, the criteria for forming random solid solutions in HEAs upon rapid solidification were determined to be $-15 < \Delta H_{\text{mix}} < 5 \text{ kJ mol}^{-1}$ and $1\% < \delta < 5\%$ [29]. On the other hand, criteria for forming a mixture of ordered solid-solution and random solid-solution are $-20 < \Delta H_{\text{mix}} < 0 \text{ kJ mol}^{-1}$ and $5\% < \delta < 6.6\%$. To further include the mixing entropy effect, an additional parameter Ω was also proposed to be useful [30]:

$$\Omega = \frac{T_m \Delta S_{\text{mix}}}{|\Delta H_{\text{mix}}|} \quad (4)$$

where T_m is the average melting temperature and ΔS_{mix} is the entropy of mixing of the alloy system, which are calculated separately as:

$$T_m = \sum_{i=1}^n c_i (T_m)_i \quad (5)$$

$$\Delta S_{\text{mix}} = -R \sum_{i=1}^n (c_i \ln c_i) \quad (6)$$

where $(T_m)_i$ stands for the melting point of the i th element and $R (= 8.314 \text{ J K}^{-1} \text{ mol}^{-1})$ is the gas constant. It was argued that Ω must be greater than unity [30]; specifically the effect of entropy of mixing must overshadow the enthalpy of mixing in order to favor formation of high-entropy phases. From available data, it was concluded that the criteria for forming single solid solutions are $\Omega \geq 1.1$ and $\delta < 6.6\%$ [30].

For the current Al-containing alloys, we plot the $\Delta H_{\text{mix}}-\delta$ and $\Omega-\delta$ graphs in Fig. 5a and b, respectively. It is evident in Fig. 5a that the $\Delta H_{\text{mix}}-\delta$ criterion might be able to predict the coexistence of ordered and disordered solid

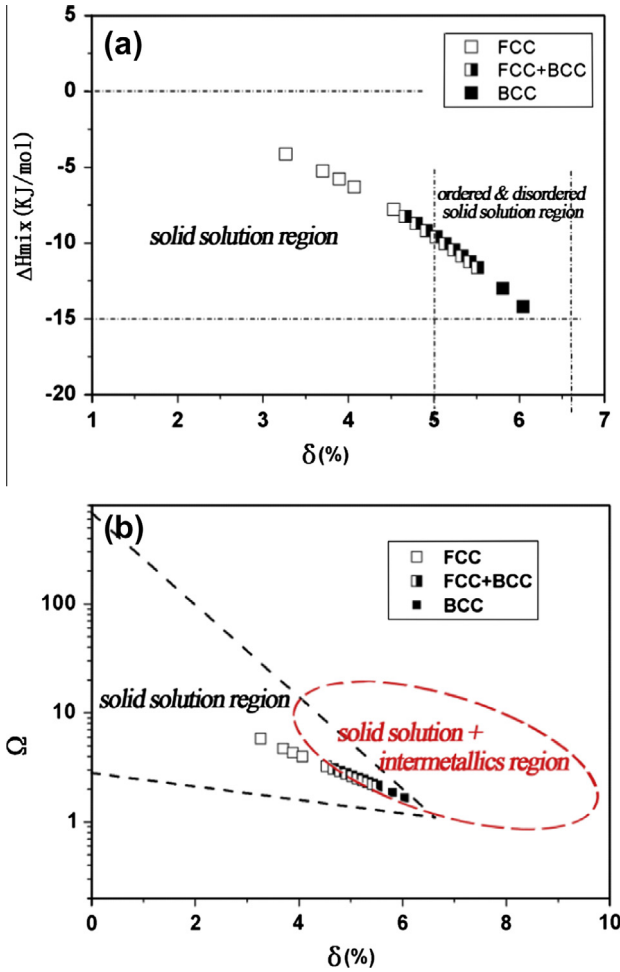


Fig. 5. Model predictions: (a) $\Delta H_{\text{mix}}-\delta$ and (b) $\Omega-\delta$ for (FeCoNiCrMn)_{100-x}Al_x alloys.

solutions, but ineffectively describes the structure transition from the single random fcc solid solution to the mixture of fcc and bcc, and to the single bcc region. For the $\Omega-\delta$ criterion, the current alloys fall into the zone where the predicted solid-solution region and solid-solution + intermetallics region overlap. Therefore, similar to the $\Delta H_{\text{mix}}-\delta$ relation, this $\Omega-\delta$ criterion reasonably predicts the formation of the solid solutions, but seems ineffectively to separate the exact crystal structure of these random solid solutions, i.e. fcc, bcc or fcc + bcc.

From an electronic structure point of view, Guo and Liu [31] and Gou et al. [32] proposed another parameter, valence electron concentration (VEC), which is essentially the e/a ratio, to predict the phase stability of HEAs:

$$\text{VEC} = \sum_{i=1}^n c_i (\text{VEC})_i \quad (7)$$

where c_i is the atomic percentage and $(\text{VEC})_i$ is the VEC for the i th element, respectively. In Guo's study, fcc solid-solution phases were found to be stable at a VEC value larger than 8, while bcc solid-solution phases were formed at a VEC value lower than 6.87. The VEC value of the current

HEA alloys is calculated and its variation with the Al content is shown in Fig. 6. It is noted that with the increase of Al, the VEC is linearly decreased, confirming that the Al addition gradually promotes covalency. Although alloys with a mixed fcc and bcc structure fall within the predicted region, the VEC criterion obviously offers a wider boundary condition for forming the single fcc and bcc solid solution phases, as compared with the experimental data. This observation suggests that in addition to atomic bonding, other factors also played an important role in phase formation resulting from the Al additions. Moreover, a more sophisticated criterion that can effectively predict the phase formation in the current HEA system and describe the alloying effect of Al is needed.

It is pointed out that aluminum atom has a larger radius than the other transition metals in the system. Thus, the solutioning of Al into FeCoNiCrMn is expected to distort the lattice. From a viewpoint of strain energy, it would cause a change of crystal structure. Fig. 7 shows the lattice constant of the FeCoNiCrMn alloys containing different amounts of Al. Evidently, as Al is initially added in the fcc HEA, the lattice constant is increased with the increase of Al, and this causes an apparent lattice straining effect [3,29]. In other words, large lattice distortion energy is produced, which destabilizes the fcc structure. Since bcc structure generally has a lower atomic packing density (68%) than fcc and hexagonal close packed structures (both 74%), it can more readily accommodate larger solute atoms like Al. Therefore, excessive addition of Al is anticipated to induce the transition from a close-packed fcc to a loose-packed bcc structure so that lattice distortion energy is relaxed [33]. This simple point of view works only if no intermetallic compound is formed.

4.2. Relationship between the structure and mechanical properties

According to Figs. 3 and 4, the mechanical properties of the current HEAs are strongly dependent upon their micro-

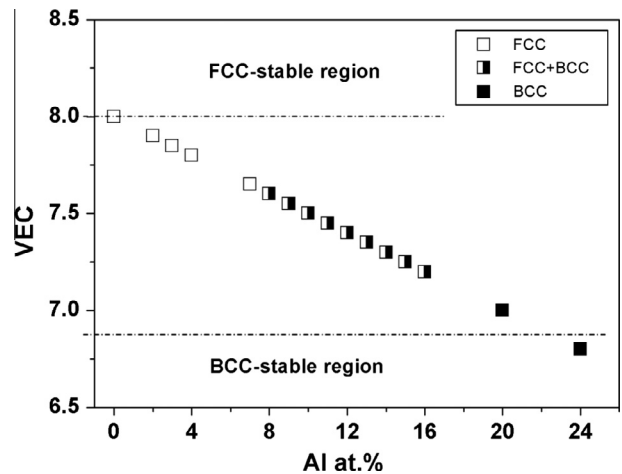


Fig. 6. The value of VEC as a function of the Al content in (FeCoNiCrMn)_{100-x}Al_x alloys.

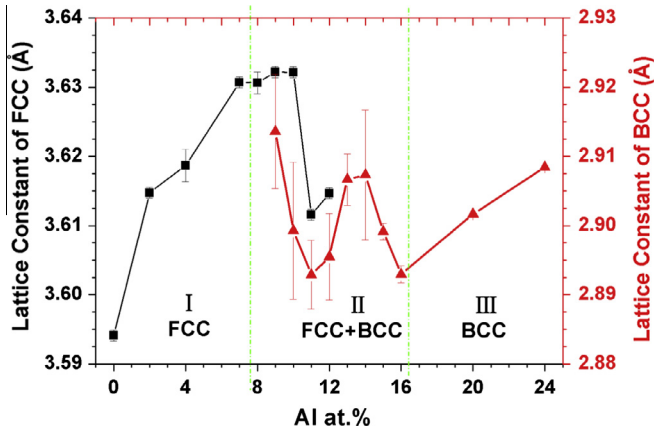


Fig. 7. Lattice constant of $(\text{FeCoNiCrMn})_{100-x}\text{Al}_x$ alloys as a function of the Al content. Solid squares and triangles represent the fcc and bcc phases, respectively.

structures. For the alloys in the single fcc solid solution phase region (i.e. region I), the lattice constant almost linearly increases with the increase of Al (see Fig. 7), which can be understood via Vegard's law [34]. The hardness, yield and tensile strength all slightly increase as the Al concentration is increased (Fig. 4), suggesting a solution strengthening effect caused by the enlarged lattice distortion; atomic radius of Al ($r_{\text{Al}} = 0.143 \text{ nm}$) is $\sim 13\%$ larger than that of Fe ($r_{\text{Fe}} = 0.126 \text{ nm}$). Here, we use the change of hardness ΔH [35] to measure the potency of solution strengthening via a standard model for substitutional solid solution strengthening based on elastic dislocation–solute interactions [36,37]:

$$\Delta H = 3^{3/2} * \frac{G * \varepsilon_s^{3/2} * c^{1/2}}{700} \quad (8)$$

where G is the shear modulus of FeCoNiCrMn, estimated to be 74 GPa [38], c is the molar ratio of Al and the $3^{3/2}$ is a conversion factor from shear stress to hardness. The interaction parameter ε_s is expressed as:

$$\varepsilon_s = \left| \frac{\varepsilon_G}{1 + 0.5\varepsilon_G} - 3 * \varepsilon_a \right| \quad (9)$$

which combines the effects of elastic and atomic size mismatch, i.e. ε_G and ε_a , and they are defined respectively as:

$$\varepsilon_G = \frac{1}{G} \frac{\partial G}{\partial c} \quad (10)$$

$$\varepsilon_a = \frac{1}{a} \frac{\partial a}{\partial c} \quad (11)$$

where a is the lattice constant of FeCoNiCrMn.

Considering that the effect of ε_G is relatively negligible in comparison with ε_a , ε_a can be obtained from Fig. 7, and thus the value of ε_s is readily calculated. For example, for alloy Al_7 , with the largest lattice constant in region I, the strength increment caused by the solution strengthening is $\sim 28 \text{ MPa}$. Compared with the experimental hardness increment between Al_0 and Al_7 ($\sim 6 \text{ HV}$ or 58.8 MPa), this increment is smaller but within the order of magnitude.

For the alloys in region II (i.e. containing 8–16% Al), the volume fraction of the bcc phases in the alloys is estimated and the results are shown in Fig. 8a. It is known that bcc phases are typically more brittle but stronger than fcc phases due to the difference in the number of their available slip systems. The fact that both the bcc volume fraction and the strength (or hardness) are approximately scaled with the Al concentration readily suggests that alloys in this region may behave like the traditional composites. Therefore, to the first approximation, the strength of the alloys can be described by the simple rule-of-mixture [39], i.e.

$$\sigma = \sigma_{\text{fcc}}^* V_{\text{fcc}} + \sigma_{\text{bcc}}^* V_{\text{bcc}} \quad (12)$$

where σ_{fcc}^* and σ_{bcc}^* are the tensile strength of the fcc and bcc phases, V_{fcc} and V_{bcc} stand for the volume fraction of the two phases, respectively, and $V_{\text{fcc}} + V_{\text{bcc}} = 1$. Eq. (12) can then be converted to

$$\sigma = \sigma_{\text{fcc}}^* + (\sigma_{\text{bcc}}^* - \sigma_{\text{fcc}}^*) V_{\text{bcc}} \quad (13)$$

Consequently, strength data measured from Al_8 to Al_{11} , i.e. alloys in region II, are plotted as a function of V_{bcc} in Fig. 8b. A good linear relationship is observed, indicating that the composite model offers reasonable interpretation for the strength increment when the Al content is within 8–11%. It is especially noted that Al_{11} alloy having a com-

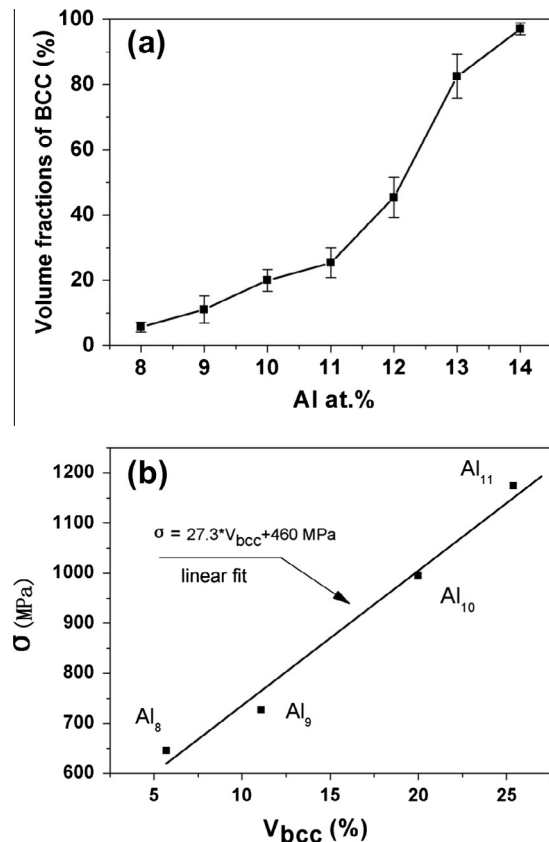


Fig. 8. (a) The volume fraction of bcc (V_{bcc}) as a function of Al measured by the grid method, and (b) the nearly linear relationship between V_{bcc} and the tensile fracture strength (σ) when the Al content is within 8–11%.

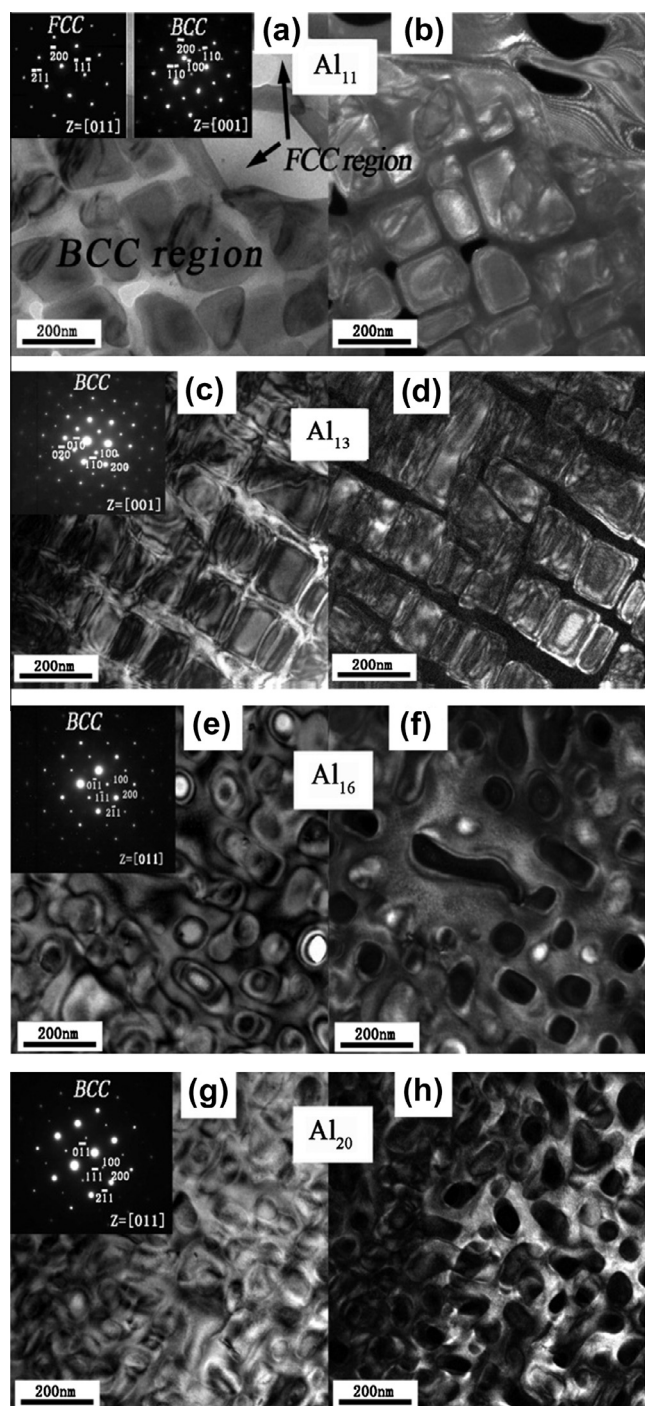


Fig. 9. TEM images of Al_8 , Al_{11} , Al_{13} , Al_{16} and Al_{20} alloys. Bright-field images are shown in (a), (c), (e), (g) and (i), respectively, while the corresponding dark-field images in the same region are shown in (b), (d), (f), (h) and (j).

posite structure exhibits the highest attainable fracture strength of 1174 MPa and yield strength of 832 MPa with a tensile ductility of 7.7%. However, alloys with more than 11% Al are too brittle to be deformed under tensile loading.

To correlate microstructure with property in the current HEA system, TEM examinations of Al_{11} , Al_{13} and Al_{16} samples have been conducted and the results are shown

in Fig. 9. Both the TEM image and the selected-area electron diffraction pattern clearly confirm that Al_{11} consists of a mixture of fcc and bcc phases. Based on the dark field image in Fig. 9b, it is revealed that the cubic-shaped particles are ordered B2 while the inter-precipitate area is disordered A2. As a matter of fact, this particular alloy is a metallic composite consisting of 74.6% fcc and 25.4% bcc phases. Such composite structure is responsible for the observed high tensile strength and good ductility. In alloy Al_{13} (Fig. 9c and d), the cubic-shaped precipitates become slightly smaller but more close-packed as compared to that in the Al_{11} alloy. In addition, the bcc volume fraction is sharply increased to over 80%. The significant increment in the ordered B2 indicates that the Al addition promotes covalency and the ordering degree in this alloy is enhanced. As the Al content is further increased, TEM images in Fig. 9e and f show that alloy Al_{16} contains spherical A2 particles with a size of 100 nm embedded in the B2 matrix. A limited fcc phase can be found only on the boundaries of the bcc grains. These observations suggest that the ordering degree is further developed, leading to an even more brittle material.

For the alloys in region III (e.g. Al_{20}), their microstructural features are similar to those in the Al_{16} alloy, as demonstrated by TEM images shown in Fig. 9g and h. The fcc particles disappear completely, and these alloys actually show a similar hardness value as that of alloy Al_{16} (see Fig. 4b), although their lattice constants increase with the Al concentration (Fig. 9).

5. Conclusions

In this study, the microstructure and tensile properties of a series of $(\text{FeCoNiCrMn})_{100-x}\text{Al}_x$ ($x = 0\text{--}20$ at.%) high-entropy alloys were characterized and evaluated. Alloying effects of Al on the phase formation, strength and ductility of these alloys were discussed. Based on obtained results and analyses, tentative conclusions can be drawn as follows.

1. From microstructural observations, the Al-containing alloys can be divided into three regions: the single fcc region with an Al concentration below 8% (region I), the duplex phases of the fcc + bcc region with the Al content between 8% and 16% (region II) and the bcc solid solution region with more than 16% Al (region III).
2. In region I ($\text{Al} < 8\%$), alloys behave like a solid solution alloy with the added Al atom as the primary strengthener. Alloys have a tensile strength of ~ 500 MPa, yield strength of ~ 220 MPa and elongation between 61.7% and 47.2%.
3. In region II ($8\% < \text{Al} < 16\%$), bcc phases begin to appear and both the fracture and yield strength are drastically increased, but the ductility is reduced sharply. Alloys in this region behave like a composite. Particularly, Al_{11} has a composite structure containing 25.4%

bcc and exhibits the highest attainable tensile strength of 1174 MPa with 7.7% ductility. However, alloys with more than 11% Al have poor ductility.

4. In region III (Al > 16%), alloys consist of disordered A2 precipitates embedded in an ordered B2 matrix. Microstructural characterization suggests that the two bcc phases were formed via spinodal decomposition. Alloys in this region are extremely brittle.
5. The observed crystalline phases in the current FeC–oNiCrMn induced by the Al addition cannot be fully predicted using the existing criteria, suggesting that additional work is necessary for the understanding of phase transition in HEAs.

Acknowledgements

This research was supported by the National Natural Science Foundation of China (Nos. 51010001, 51001009 and 51271212), 111 Project (B07003) and Program for Changjiang Scholars and Innovative Research Team in University. X.J.L. acknowledges the financial support from the Beijing Nova Program of China (No. 2010B017). The financial support for X.J.L. and Y.W. from the Fundamental Research Funds for the Central Universities is acknowledged. T.G.N. was supported by the National Science Foundation under Contract DMR-0905979.

References

- [1] Yeh JW, Chen YL, Lin SJ, Chen SK. *Mater Sci Forum* 2007;560:9.
- [2] Yeh JW. *Ann Chim Sci Mater* 2006;31:633.
- [3] Yeh JW, Chen SK, Lin SJ, Gan JY, Chin TS, Shun TT, et al. *Adv Eng Mater* 2004;6:299.
- [4] Senkov ON, Wilks GB, Miracle DB, Chuang CP, Liaw PK. *Intermetallics* 2010;18:1758.
- [5] Liu WH, Wu Y, He JY, Nieh TG, Lu ZP. *Scripta Mater* 2013;68(7):526–9.
- [6] Cantor B, Chang I, Knight P, Vincent A. *Mater Sci Eng A* 2004;375:213.
- [7] Zhang KB, Fu ZY. *Intermetallics* 2012;22:24–32.
- [8] Ma SG, Zhang Y. *Mater Sci Eng A* 2012;532:480–6.
- [9] Li BS, Wang YP, Ren MX, Yang C, Fu HZ. *Mater Sci Eng A* 2008;498(1–2):482–6.
- [10] Zhou YJ, Zhang Y, Wang YL, Chen GL. *Appl Phys Lett* 2007;90(18):181904.
- [11] Shun TT, Chang LY, Shiu MH. *Mater Sci Eng A* 2012;556:170–4.
- [12] Li C, Li JC, Zhao M, Jiang Q. *J Alloys Compd* 2010;504:S515–8.
- [13] Hsu US, Hung UD, Yeh JW, Chen SK, Huang YS, Yang CC. *Mater Sci Eng A* 2007;460–461:403–8.
- [14] Hu ZH, Zhan YZ, Zhang GH, She J, Li CH. *Mater Des* 2010;31(3):1599–602.
- [15] Yang X, Zhang Y, Liaw PK. *Proc Eng* 2012;36:292–8.
- [16] Otto F, Yang Y, Bei H, George EP. *Acta Mater* 2013;61:2628–38.
- [17] Larn RH, Yang JR. *Mater Sci Eng A* 1999;264:139–50.
- [18] Laughlin DE, Soffa WA. *Metals handbook*, vol. 9, 9th ed. Materials Park (OH): ASM; 1985.
- [19] Para S. *Metallography and microstructure*. ASM handbook, vol. 9. Materials Park (OH): ASM, International; 2004.
- [20] Wu JM, Lin SJ, Yeh JW, Chen SK, Huang YS, Chen HC. *Wear* 2006;261(5–6):513–9.
- [21] Wang WR, Wang WL, Wang SC, Tsai YC, Lai CH, Yeh JW. *Intermetallics* 2012;26:44–51.
- [22] Boyer HE, Gall TL. *Metals handbook*. Desk ed. Materials Park (OH): ASM International; 1985.
- [23] Tabor D. *Hardness of metals*. Oxford: Clarendon Press; 1951.
- [24] Hollomon JH. *Trans AIME* 1945;162:268–90.
- [25] Dieter GE. *Mechanical metallurgy*. 3rd ed. New York: McGraw-Hill; 1988.
- [26] Honeycombe RWK, Bhadeshia HKDH. *Steels: microstructure and properties*. 3rd ed. Oxford: Elsevier; 2006.
- [27] Tung CC, Yeh JW, Shun TT, Chen SK, Huang YS, Chen HC. *Mater Lett* 2007;61:1.
- [28] Zhang Y, Zhou YJ, Lin JP, Chen GL, Liaw PK. *Adv Eng Mater* 2008;10(6):534–8.
- [29] Egami T, Waseda YJ. *Non-Cryst Solids* 1984;64:113.
- [30] Yang X, Zhang Y. *Mater Chem Phys* 2012;132(2–3):233–8.
- [31] Guo S, Liu CT. *Prog Nat Sci: Mater Int* 2011;21(6):433–46.
- [32] Guo S, Ng C, Lu J, Liu CT. *J Appl Phys* 2011;109(10):103505.
- [33] Zhang Y, Chen GL, Gan CL. *J ASTM Int* 2010;7(5):1–8.
- [34] Somdutta M, Vishal R, Rajeev G, Ashish G. *Solid State Commun* 2012;152(13):1181–5.
- [35] Schuh CA, Nieh TG, Iwasaki H. *Acta Mater* 2003;51(2):431–43.
- [36] Fleischer RL. *Acta Metall* 1963;11:203.
- [37] Courtney TH. *Mechanical behavior of materials*. New York: McGraw-Hill; 1990.
- [38] Zhu C, Lu ZP, Nieh TG. *Acta Mater* 2013;61(8):2993–3001.
- [39] Tsai SW. *Composites design*. Dayton (OH): Think Composites Press; 1988.